SCHOLAREASE – INTELLIGENT SCHOLARSHIP APPLICATION, ELIGIBILITY AND FINANCIAL AID MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

VISHAALI R

(2116240701595)

in partial fulfilment of the End Semester Examination for the course

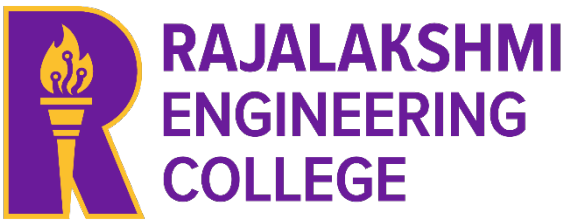
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



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(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "ScholarEase – Intelligent Scholarship Application, Eligibility and Financial Aid Management System" is the bonafide work of VISHAALI R (2116240701595), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

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DECLARATION

I, VISHAALI R (2116240701595), hereby declare that the project report titled "ScholarEase – Intelligent Scholarship Application, Eligibility and Financial Aid Management System", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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VISHAALI R

ABSTRACT

Scholarship administration in engineering colleges is still largely handled through paper application forms, photocopied certificates, registers and spreadsheets maintained by office staff. Every application has to be checked by hand against the CGPA limit, the family-income ceiling and the deadline of the chosen scheme, certificate details have to be read and retyped, and decisions are communicated through notices or personal contact. The process is slow, difficult to track and prone to inconsistent eligibility decisions. ScholarEase – Intelligent Scholarship Application, Eligibility and Financial Aid Management System was developed on the Zoho Creator low-code platform to centralise and automate this process. A low-code approach was chosen because the problem is data- and workflow-centric, and Zoho Creator provides hosted forms, a relational data store, reports, pages, Deluge scripting, built-in AI fields and external connections without separate infrastructure. The application contains five forms (Student Details, Scholarship Details, Scholarship Application, Application Review and AI Scholarship Recommendations) linked through lookup fields, ten list reports, a home page and a dashboard page with KPI cards and charts. Ten form workflows and six Deluge custom functions implement automatic eligibility verification against CGPA, income and deadline rules, review-status propagation, e-mail notifications, and an OpenAI Responses API integration through a secured connection for advisory scholarship recommendations; a Zia OCR AI field extracts visible text from uploaded certificate images. Sixteen test cases were executed on the live application on 01 November 2026. Student and scholarship registration, mandatory-field validation, eligible, ineligible and deadline-based eligibility decisions, OCR extraction, all three review decisions, the submission and decision e-mails, report filtering and dashboard KPI updates behaved as expected; the OpenAI recommendation step failed because the linked OpenAI account had no remaining API credits. The application is currently maintained in the development environment and has not been published to the Stage or Production environments. The system removes manual eligibility calculation, gives administrators a single, up-to-date view of every application and keeps final approval with human reviewers.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Scholarship Management; Optical Character Recognition*

PROJECT AT A GLANCE

Attribute	Details
Project Title	ScholarEase – Intelligent Scholarship Application, Eligibility and Financial Aid Management System
Application Domain	Education – scholarship and financial aid administration
Student Name & Register No.	Vishaali R – 2116240701595
Zoho Creator Edition	Zoho Creator free trial of a paid plan (trial active at the time of writing)
Application Name (Link Name)	ScholarEase 1 (scholarease-1)
Published / Live URL	https://creatorapp.zoho.in/vishaali.r.2024.cse_rajal188/scholarease-1 (development environment; not published to Production)
Demo Credentials (per role)	Single administrator account (application owner). No additional demo users or roles have been created.
Number of Forms / Subforms	5 / 0
Number of Reports (by type)	List: 10 Kanban: 0 Calendar: 0 Pivot/Chart: 0 Others: 0
Number of Pages / Dashboards	2 (ScholarEase Home, Scholarship Dashboard.)
Number of Workflows	Form: 10 (8 with actions, 2 empty) Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	6 custom functions / 0 custom report actions
Integrations	OpenAI Responses API (connection openai_connection); Zia OCR AI field; Zoho Creator sendmail (e-mail)
Roles & Profiles Defined	Application owner (administrator); default Read and Write permission profiles only
Environments Used	Development only (Stage and Production not used)
Date of Production Deployment	Not deployed to Production

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CO	Course Outcome
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
ERD	Entity–Relationship Diagram
FR	Functional Requirement
HTTP	Hypertext Transfer Protocol
ID	Identifier
INR	Indian Rupee
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
OAuth	Open Authorization
OCR	Optical Character Recognition
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho's AI assistant and AI services

CHAPTER 1
INTRODUCTION

1.1 Background

Scholarships help academically capable students continue their studies irrespective of their economic background. In a typical engineering college, students are eligible for several categories of support at the same time: merit-based scholarships that reward academic performance, need-based scholarships that depend on the annual family income, and academic-excellence awards offered by trusts, government departments and private foundations. Each scheme publishes its own eligibility conditions, most commonly a minimum CGPA, a maximum annual family income and an application deadline, together with the amount that is awarded.

ScholarEase has been developed for this scholarship-administration process in a college setting. It is built as a Zoho Creator application in which students, scholarships, applications, review decisions and AI-generated recommendations are stored as related records, and in which eligibility checking, status updates and notifications are carried out automatically by Deluge scripts and workflows.

1.2 Problem Statement

Scholarship applications are currently collected, verified and tracked manually, which makes the process slow, inconsistent and difficult to audit. The specific pain points are:

- Student academic and financial information is collected repeatedly on paper and maintained in separate registers or spreadsheets.
- Eligibility for multiple scholarships with different CGPA, income and deadline rules has to be checked manually, which is time-consuming and error-prone.
- Details printed on certificates have to be read and typed in by hand during verification.
- Students find it difficult to identify which scholarships suit their academic and financial profile.
- There is no centralised record of application and review status, and decisions reach students late.
- Statistics such as the number of eligible, approved and rejected applications are compiled manually.

1.3 Objectives

The objectives of the project are:

- To design a normalised data model and Zoho Creator forms for students, scholarships, applications, reviews and AI recommendations, linked through lookup fields.
- To automate scholarship eligibility verification against minimum CGPA, maximum family income and application deadline using Deluge scripting and form workflows.
- To automate the propagation of administrative review decisions to the application record and to notify students by e-mail.
- To integrate the application with the OpenAI Responses API through a secured connection for advisory scholarship recommendations, and to use a Zia OCR AI field to extract text from certificate images.
- To provide administrators with filtered reports and a dashboard of KPI cards and charts computed from live application records.
- To prepare the application for secure deployment to a production environment with governance controls.

1.4 Scope of the Project

In scope. The project covers the complete scholarship cycle inside a single Zoho Creator application: registration of student academic and financial details; creation and maintenance of scholarship schemes with their eligibility criteria; submission of applications with a supporting document and a certificate image; automatic eligibility evaluation; OCR-based extraction of certificate text; AI-based scholarship recommendations on request; recording of review decisions with automatic status updates; e-mail notifications; ten list reports; a home page; and a dashboard page. The application is accessed through the Zoho Creator web interface, and the same forms are available in the Zoho Creator mobile layout.

Out of scope. Online disbursement of scholarship amounts, integration with government scholarship portals or the college ERP, verification of certificate authenticity, and a separate student self-service portal with restricted role-based access are not part of the current implementation. AI recommendations are advisory only; the

application does not approve or reject scholarships automatically, and final decisions are always recorded by an administrator.

1.5 Justification for a Low-Code Approach

The scholarship problem is primarily about capturing structured data, relating records to one another, applying rule-based decisions and presenting the results to administrators. These are exactly the capabilities that low-code application platforms provide out of the box [1], [2]. Zoho Creator supplies a hosted relational data store, drag-and-drop form and report builders, a page builder for dashboards, the Deluge scripting language for business logic, workflow triggers on form events, built-in Zia AI fields such as OCR, and connections for calling external REST APIs securely [3], [4]. Hosting, authentication, backups and the mobile layout are handled by the platform, so the development effort could be spent on the scholarship rules themselves rather than on infrastructure. Research on low-code platforms identifies faster delivery and the involvement of domain users as their main advantages, together with platform dependence as the main risk [5], [6]; these trade-offs are discussed in Chapter 8.

1.6 Organisation of the Report

Chapters 2 and 3 cover the system study and requirements, Chapters 4 and 5 the design and implementation, Chapter 6 testing, Chapter 7 deployment, security and governance, Chapter 8 results and discussion, and Chapter 9 the conclusion and future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing manual process, the scholarship cycle starts when a notice for a scheme is circulated. Interested students collect a paper form, fill in their personal, academic and income details, and attach photocopies of their mark sheets and income certificate. The forms are submitted to the office, where staff enter the key details into a register or spreadsheet. For each form, staff look up the rules of the chosen scheme and compare the student's CGPA, annual family income and submission date with the minimum CGPA, maximum income and deadline. Certificate details are read and typed by hand. The files are then reviewed by a committee, the decision is noted against each application, and students are informed through notices or personal communication. At the end of the cycle, staff count eligible, approved and rejected applications manually to prepare reports. Figure 2.1 shows this as-is flow.

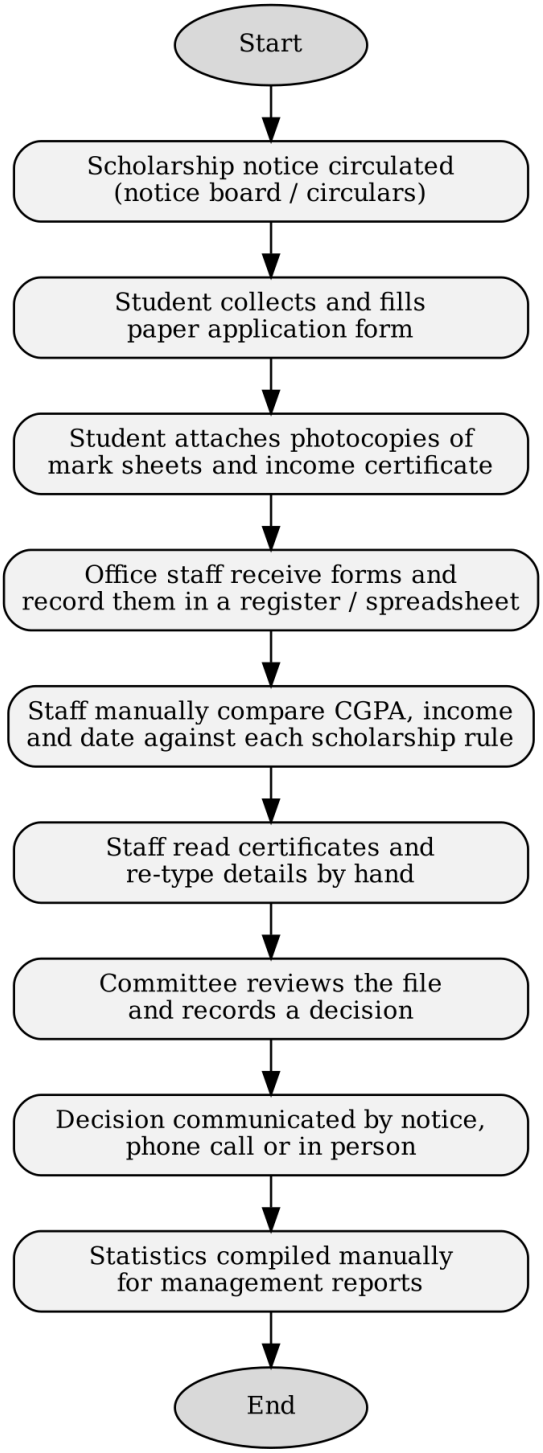


Figure 2.1: As-Is Process Flow of the Existing Manual Scholarship System

2.2 Limitations of the Existing System

- Repeated manual data entry of the same student information across registers and spreadsheets leads to duplication and transcription errors.
- Eligibility checks for several schemes with different thresholds are performed by hand, so two staff members may reach different conclusions for the same case.

- Certificate details have to be read and retyped, which slows down verification.
- Students have no single place to discover suitable schemes or to see the status of their applications.
- Decisions are communicated late and inconsistently because there is no automated notification.
- Management statistics are produced by manual counting, and there is no real-time view of applications by status, month or department.

2.3 Low-Code versus Traditional Development

Table 2.1 compares a conventional pro-code implementation of ScholarEase (for example, a web front end, a back-end framework and a relational database) with the Zoho Creator implementation that was actually built.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Several weeks to design the database schema, build the back-end APIs, the front-end screens, the authentication module and the charts.	The five forms, ten reports, two pages and ten workflows were built within a single semester project timeline using visual builders and short Deluge scripts.
Skills required	Front-end (HTML, CSS, JavaScript), back-end language and framework, SQL, deployment and security skills.	Knowledge of the Zoho Creator builders and Deluge scripting; basic REST/JSON understanding for the OpenAI connection.
Hosting & infrastructure	Separate web server, database server, SSL certificates, backups and monitoring must be arranged.	Hosted by Zoho; the application is available at a creatorapp.zoho.in URL without any server setup.
Mobile application	A responsive design or a separate mobile application has to be developed.	Forms and reports automatically render in the Zoho Creator mobile layout (Figure 5.9).
Security & access control	Authentication, sessions, password storage and role checks must be coded and tested.	Zoho account authentication, permission profiles, private fields and connections are provided by the platform.
Maintenance & change	Schema migrations and code changes require redeployment.	Fields, reports and workflows are changed visually; scripts are edited in the Deluge editor or the Application IDE.
Customisation limits	Unlimited control over user interface and logic.	User interface is limited to builder components and page snippets; logic is limited to Deluge and platform quotas.
Cost	Developer time plus hosting and maintenance costs.	Subscription cost per user after the trial period; no separate hosting cost.

2.4 Literature Survey

The literature survey covered research on low-code development platforms, the documentation of Zoho Creator, work on optical character recognition, and an existing scholarship management product. Table 2.2 summarises the sources.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020 [7]	Supporting the understanding and comparison of low-code development platforms	Defines common features of low-code platforms (graphical modelling, data modelling, workflow, integration, deployment) and compares major platforms.	Used to identify which Zoho Creator features (forms, reports, workflows, connections) map to the scholarship requirements.
2	A. C. Bock and U. Frank, 2021 [5]	Low-Code Platform	Characterises low-code platforms as model-driven environments that shift development towards domain users, with risks of vendor lock-in.	Supports the justification for a low-code approach and the discussion of limitations in Chapter 8.
3	D. Di Ruscio et al., 2022 [6]	Low-code development and model-driven engineering: Two sides of the same coin?	Relates low-code platforms to model-driven engineering and discusses abstraction, scalability and interoperability concerns.	Informs the reflection on the no-code to pro-code spectrum in Section 8.5.
4	R. Smith, 2007 [8]	An Overview of the Tesseract OCR Engine	Describes the stages of an OCR engine (layout analysis, line and word recognition) and how recognition quality depends on image quality.	Explains why OCR output must be checked by an administrator and why image quality affects extraction accuracy.
5	Zoho Corporation [2], [3]	Zoho Creator Help Documentation and Deluge Reference	Documents forms, lookup fields, reports, pages, form workflows, AI fields, connections and Deluge syntax such as fetch, sendmail and invokeurl.	Primary implementation reference for every module of ScholarEase.
6	Government of India [9]	National Scholarship Portal	Provides a single online portal for central and state scholarship schemes with online application and status tracking.	Demonstrates the value of centralised online scholarship processing; ScholarEase applies the same idea at institution level.

2.5 Emerging Trends in Low-Code Development

Three trends in low-code development influenced the design of ScholarEase. First, **AI-assisted development and AI services** are now built into low-code platforms. Zoho Creator offers Zia AI fields such as OCR that can be added to a form like any other field, and ScholarEase uses this feature to extract certificate text without writing image-processing code. Large language models can be called from the platform; ScholarEase calls the OpenAI Responses API through a connection to generate advisory recommendations. Second, **citizen development** encourages domain users to build and maintain their own applications. The forms, reports and dashboard of ScholarEase can

be changed visually by administrative staff when a new scheme or field is required. Third, **hyper-automation** combines rules, workflows and AI to remove manual steps; in ScholarEase, eligibility checking, status propagation and notifications run automatically when a record is submitted, while human reviewers remain responsible for final decisions. Industry analyses of enterprise low-code platforms describe the same direction [10].

2.6 Proposed System

The proposed system, ScholarEase, replaces the manual process with a single Zoho Creator application. Student details and scholarship schemes are stored once and referenced through lookup fields, which removes repeated data entry. Eligibility is evaluated automatically by a Deluge script when an application is submitted, which makes decisions consistent. Certificate images are processed by a Zia OCR AI field so that visible text is available for verification without retyping. Students can request AI-based recommendations that compare their profile with the stored schemes. Review decisions are recorded in a separate form and automatically update the application status, and e-mail workflows notify students of submissions and decisions. Filtered reports and a dashboard give administrators a real-time picture of applications by eligibility, review status, month and department. Each limitation listed in Section 2.2 is therefore addressed by a specific module described in Chapter 5.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Scholarship cell staff member who owns and manages the application. In the current build this is the application owner account.	Create and maintain scholarship schemes, review applications and record decisions, verify certificate text, monitor reports and the dashboard, manage workflows and settings.
Student	Applicant who wishes to apply for scholarships.	Register academic and financial details, view scholarships, submit applications with documents, request AI recommendations, receive e-mail notifications.
External services	OpenAI Responses API and Zoho platform services (Zia OCR, e-mail).	Generate recommendation text, extract certificate text and deliver e-mail messages.

In the current implementation all forms are operated through the administrator (owner) account; the student role is represented by the data entered on behalf of the student and by the e-mail address recorded in the Student Details form. Separate student user accounts are listed as a future enhancement in Section 9.2.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow a student to be registered with Student Name, Student ID, Email, Department, Year of Study, CGPA and Annual Family Income, all mandatory.	Administrator / Student	High
FR-02	The system shall allow an administrator to create scholarships with name, type, minimum CGPA, maximum family income, amount and application deadline.	Administrator	High
FR-03	The system shall allow an application to be submitted that links a student and a scholarship, with an optional supporting document, a certificate image and an application date.	Student / Administrator	High
FR-04	The system shall automatically set Eligibility Status to Eligible only when CGPA ≥ minimum CGPA, income ≤ maximum income and application date ≤ deadline, and to Not Eligible otherwise.	System	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall extract visible text from an uploaded certificate image into the Extracted Certificate Text field.	System	Medium
FR-06	The system shall send the student profile, the request text and the stored scholarships to the OpenAI Responses API and store the returned recommendation.	Student / System	Medium
FR-07	The system shall update the Review Status of the linked application to Approved, Rejected or Additional Documents Required when a review decision is submitted.	Administrator / System	High
FR-08	The system shall send an e-mail confirmation when an application is submitted.	System	Medium
FR-09	The system shall send an e-mail to the student when a review decision is recorded.	System	Medium
FR-10	The system shall display KPI cards and charts of total, eligible, approved and rejected applications, monthly trends and department-wise counts.	Administrator	Medium
FR-11	The system shall provide reports filtered by eligibility and review status.	Administrator	Medium
FR-12	The system shall provide a home page with navigation to the main modules.	All users	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and list reports with the current data volume (tens of records) shall load within a few seconds in a standard browser.
NFR-02	Security	Only authenticated Zoho users shall access the application; API credentials shall be stored in a Zoho connection and not in scripts; status and AI fields shall be private.
NFR-03	Usability	The application shall be usable on the Zoho Creator web interface and the mobile layout, with clear field labels and success messages.
NFR-04	Availability	The application shall be available whenever the Zoho Creator service is available, without institution-managed servers.
NFR-05	Maintainability	Business rules shall be implemented in named workflows and custom functions that can be reviewed in the Application IDE.
NFR-06	Auditability	Each application shall retain its eligibility and review status, and each review decision shall be stored as a separate record with reviewer and date.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (free trial of a paid plan), data centre zoho.in
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing	Deluge info statements and the form "View Log Details" output (Section 6.3)
Other services	Zia OCR AI field; OpenAI Responses API (model gpt-4.1-mini) through a Zoho connection; Zoho Creator sendmail
Mobile	Zoho Creator mobile layout (preview used for verification)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases of ScholarEase for the two human actors. Submission of an application always includes the automatic eligibility check. Table 3.5 describes the main use case in detail.

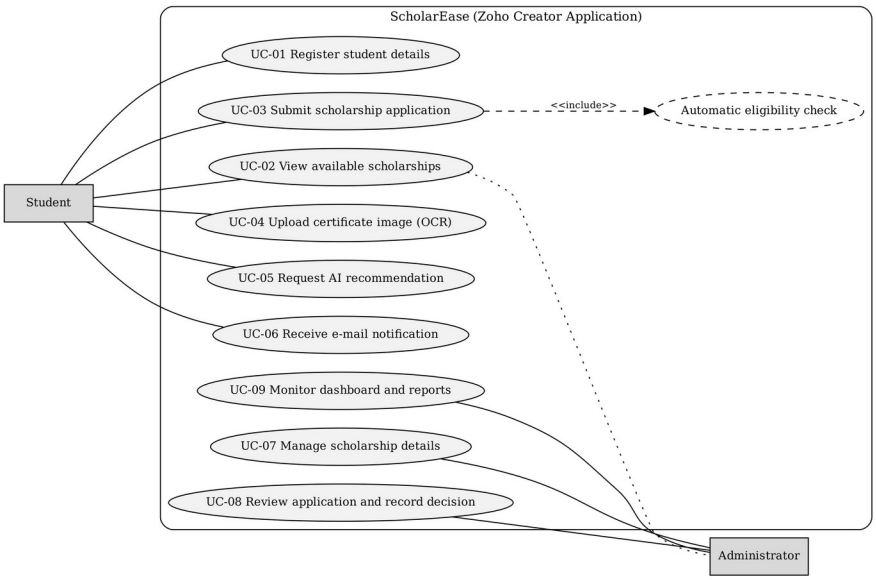


Figure 3.1: Use Case Diagram of ScholarEase

Table 3.5: Use Case Description – Submit Scholarship Application

Field	Description
Use Case ID	UC-03
Use Case Name	Submit scholarship application
Actor(s)	Student (primary); System (eligibility workflow, OCR field, e-mail workflow)

Field	Description
Pre-condition	The student is registered in Form 1: Student Details and the scholarship exists in Scholarship Details.
Main Flow	1. Open the Scholarship Application form. 2. Select the scholarship and the student record. 3. Upload the certificate image (optional supporting document). 4. Confirm the application date and submit. 5. The On Validate workflow computes Eligibility Status. 6. The record is saved and the success message is shown. 7. On Success workflows send the confirmation e-mail.
Alternate Flow	If the mandatory Application Date is missing, the form shows a validation message and the record is not saved. If the student or scholarship is not selected, the eligibility script is skipped.
Post-condition	A Scholarship Application record exists with an Application ID, Eligibility Status and (if an image was uploaded) Extracted Certificate Text.
Related FR	FR-03, FR-04, FR-05, FR-08

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

ScholarEase follows a layered architecture, shown in Figure 4.1. All layers run inside the Zoho Creator platform except the OpenAI Responses API, which is an external service.

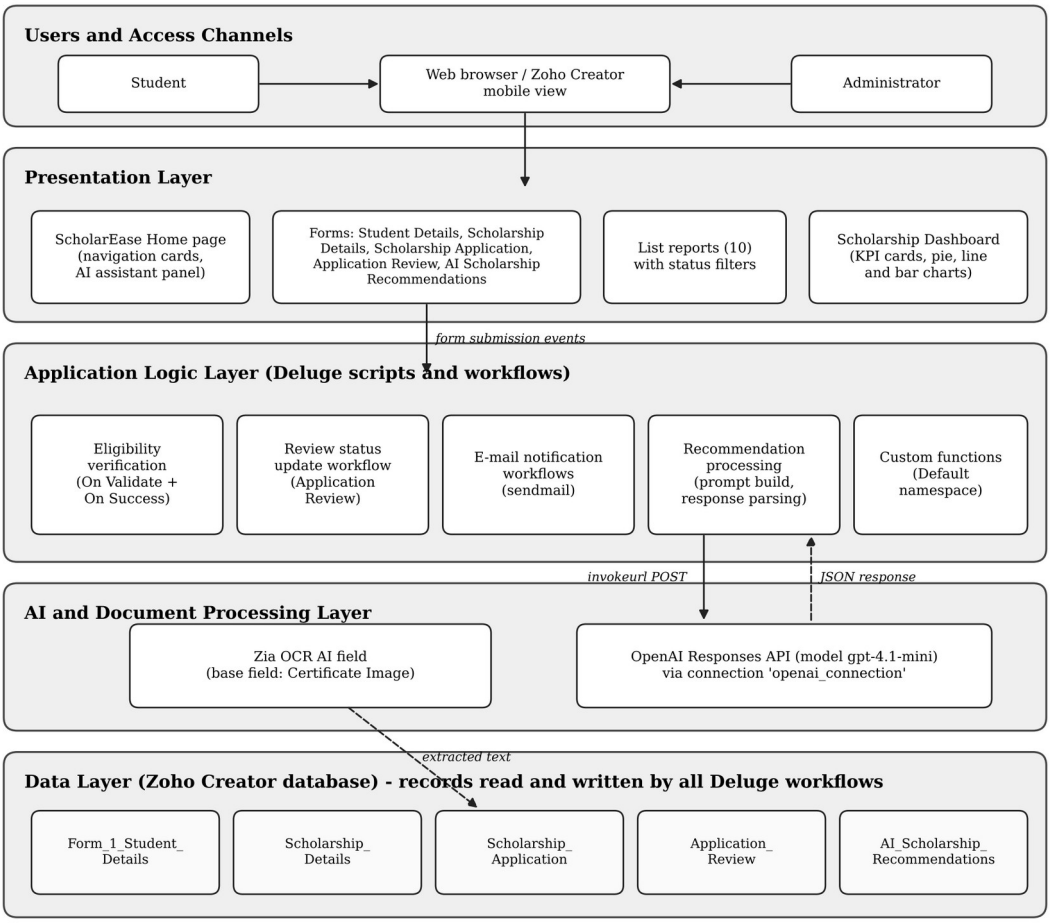


Figure 4.1: System Architecture of ScholarEase

Users and access channels. Students and the administrator use the application through a web browser at its creatorapp.zoho.in address; the same components are rendered in the Zoho Creator mobile layout.

Presentation layer. The ScholarEase Home page provides navigation cards, the five forms capture data, ten list reports display filtered records, and the Scholarship Dashboard page shows KPI cards and charts.

Application logic layer. Deluge scripts attached to form events implement the business rules: eligibility verification on the Scholarship Application form, review-status propagation and decision e-mails on the Application Review form, submission e-mails, recommendation processing on the AI Scholarship Recommendations form, and six reusable custom functions.

AI and document processing layer. The Zia OCR AI field reads the uploaded certificate image and fills the Extracted Certificate Text field. The recommendation workflow sends a JSON request to the OpenAI Responses API through the connection openai_connection and parses the JSON response.

Data layer. Five Zoho Creator forms act as tables. Records flow upward to reports and dashboard aggregates, and downward when workflows create or update records.

Information flow. When a user submits a form, the presentation layer passes the input to the On Validate and On Success events in the logic layer. Scripts fetch related records from the data layer, apply rules, and either set field values before the record is saved or update related records afterwards. External calls go to the AI layer and their responses are written back to the data layer, from where reports and the dashboard read them.

4.2 Data Model

The data model contains five entities, each implemented as a Zoho Creator form. Relationships are realised through lookup fields; no subforms are used. Figure 4.2 shows the entity–relationship diagram generated from the actual field definitions; the Schema Builder view of the same relationships is shown in Figure 5.2.

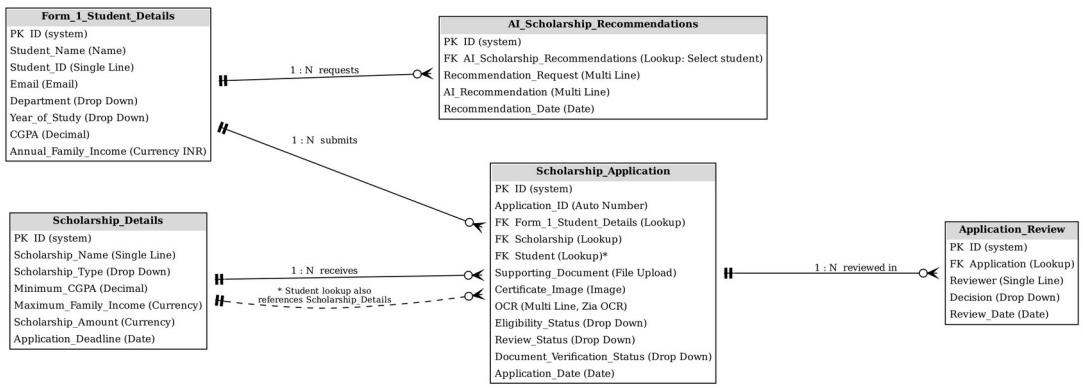


Figure 4.2: Entity–Relationship Diagram of ScholarEase

One student can submit many applications and many recommendation requests; one scholarship can receive many applications; and one application can have one or more

review records, the latest of which determines its Review Status. The Scholarship_Application form also contains a field labelled Student whose lookup points to Scholarship_Details; the student relationship used by all scripts and reports is the Form_1_Student_Details lookup.

4.3 Form Design

Tables 4.1 to 4.5 give the field specification of every form exactly as configured in the application (link names are taken from the Application IDE).

Table 4.1: Field Specification – Form 1: Student Details

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Name	Yes	First and last name shown; prefix and suffix hidden	Marked as personal data
Student ID	Student_ID	Single Line	Yes	—	College identifier
Email	Email	Email	Yes	E-mail format; max 80 characters	Personal data; used for notifications
Department	Department	Drop Down	Yes	CSE, ECE, IT, EEE, CIVIL, CSBS; default ECE	Used in department chart
Year of Study	Year_of_Study	Drop Down	Yes	1st year to 4th year	—
CGPA	CGPA	Decimal	Yes	Numeric	Compared with Minimum CGPA
Annual Family Income	Annual_Family_Income	Currency (INR)	Yes	Indian comma format	Compared with Maximum Family Income

Table 4.2: Field Specification – Scholarship Details

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Scholarship Name	Scholarship_Name	Single Line	Yes	—	Display value in lookups
Scholarship Type	Scholarship_Type	Drop Down	No	Merit-Based, Need-Based, Academic Excellence	—
Minimum CGPA	Minimum_CGPA	Decimal	Yes	Numeric	Eligibility threshold
Maximum Family Income	Maximum_Family_Income	Currency (INR)	Yes	0 decimal places	Eligibility threshold
Scholarship Amount	Scholarship_Amount	Currency (INR)	Yes	0 decimal places	—
Application Deadline	Application_Deadline	Date	Yes	All weekdays allowed	Eligibility threshold

Table 4.3: Field Specification – Scholarship Application

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Application ID	Application_ID	Auto Number	—	System generated	Shown in review lookup
Student	Student	Lookup (Scholarship_Details)	No	—	Secondary lookup; not used by scripts
Scholarship	Scholarship	Lookup (Scholarship_Details)	No	Displays Scholarship_Name	Used by eligibility script
Supporting Document	Supporting_Document	File Upload	No	—	Stores PDF or other documents
Certificate Image	Certificate_Image	Image	No	—	Base field for OCR
Eligibility Status	Eligibility_Status	Drop Down	No	Pending Check, Eligible, Not Eligible	Private; set by script
Review Status	Review_Status	Drop Down	No	Pending, Under Review, Approved, Rejected, Additional Documents Required	Private; set by review workflow
Application Date	Application_Date	Date	Yes	Initial value 13-Oct-2026	Compared with deadline
Form 1: Student Details	Form_1_Student_Details	Lookup (Form_1_Student_Details)	No	Displays first and last name	Student link used by scripts
Extracted Certificate Text	OCR	Multi Line (Zia OCR AI field)	No	Base field Certificate_Image	Private; filled by OCR
Document Verification Status	Document_Verification_Status	Drop Down	No	pending, verified, rejected	Private; set by administrator

Table 4.4: Field Specification – Application Review

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Application	Application	Lookup (Scholarship_Application)	No	Displays Application_ID	Links decision to application
Reviewer	Reviewer	Single Line	Yes	—	Name of reviewer
Decision	Decision	Drop Down	Yes	Approved, Rejected, Additional Document Needed	Drives status and e-mail workflows
Review Date	Review_Date	Date	No	All weekdays allowed	—

Table 4.5: Field Specification – AI Scholarship Recommendations

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Select student	AI_Scholarship_Recommendations	Lookup (Form_1_Student_Details)	No	Displays full name	Student whose profile is used
Recommendation Request	Recommendation_Request	Multi Line	Yes	Tooltip guides the user	Free-text question

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
AI Recommendation	AI_Recommendation	Multi Line	No	—	Private; written by workflow
Recommendation Date	Recommendation_Date	Date	No	—	Written by workflow

4.4 Process and Workflow Design

Figure 4.3 shows the complete process from student registration to dashboard update, including the decision made by the eligibility script and the points at which On Validate and On Success workflows run.

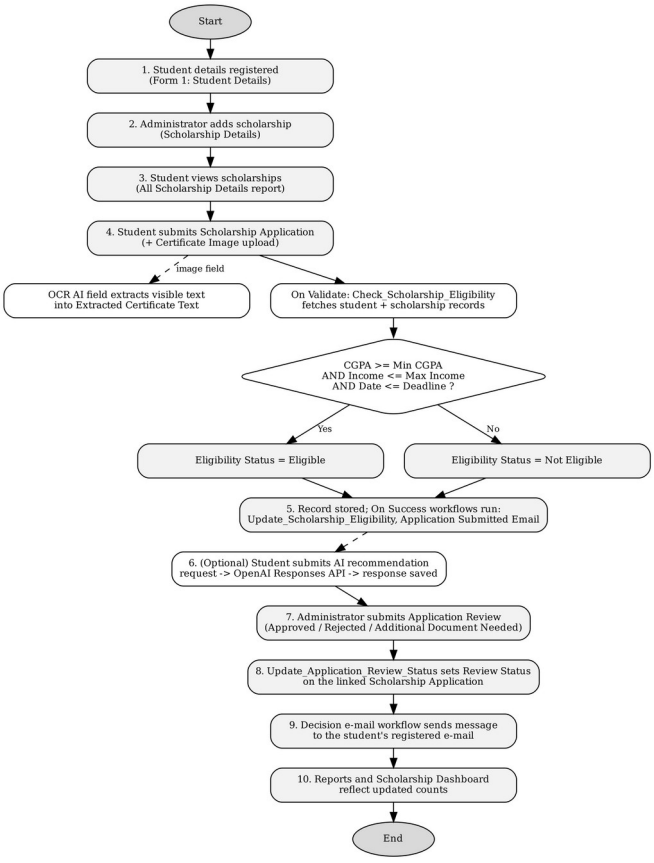


Figure 4.3: Workflow / Process Flow for the Complete Scholarship Application Cycle

4.5 User Interface and Navigation Design

The navigation of the application is organised into sections in the left menu, each grouping a form with its reports. Figure 4.4 shows the navigation structure as configured.

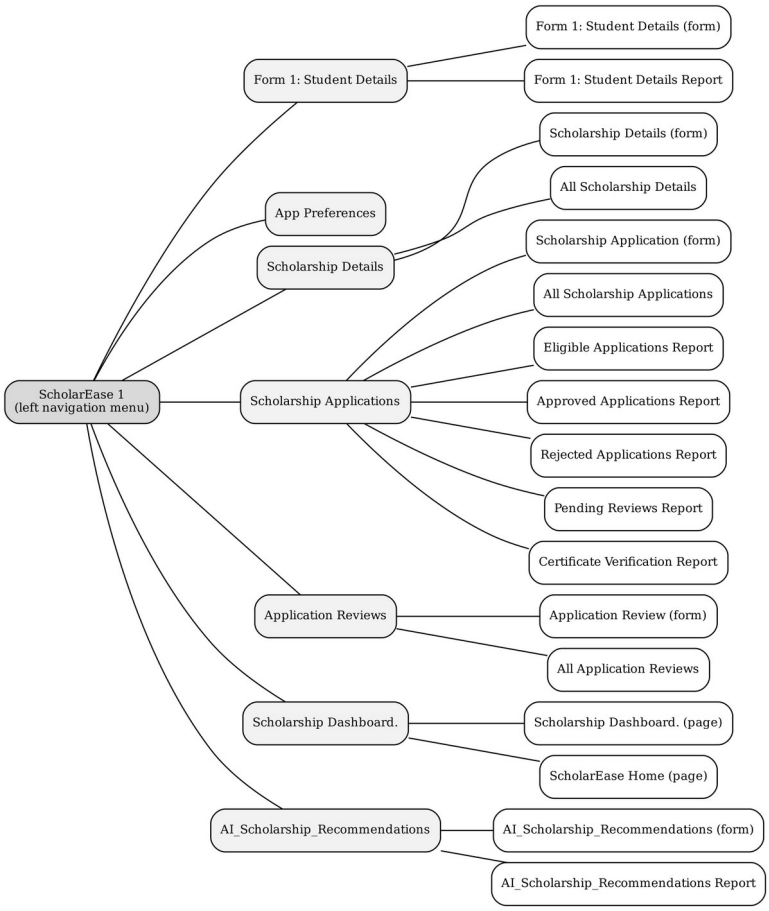


Figure 4.4: Navigation Structure of ScholarEase

The ScholarEase Home page is designed as a landing page with a title banner, a welcome message, four navigation cards (Open Student Details, View Scholarships, Apply Now, Check Status), a dashboard card and an AI assistant panel. The Scholarship Dashboard page is designed with four KPI cards on top, followed by two pie charts, a second KPI strip, a line chart, a bar chart and an embedded report of all applications.

4.6 Access Control Design

Table 4.6 gives the intended permission matrix (A = Add, E = Edit, V = View, D = Delete, — = No access). Section 5.8 explains which parts of this matrix are implemented in the current build.

Table 4.6: Role–Permission Matrix

Component	Administrator	Student (planned)
Form: Student Details	A E V D	A V (own)
Form: Scholarship Details	A E V D	—
Form: Scholarship Application	A E V D	A V (own)
Form: Application Review	A E V D	—

Component	Administrator	Student (planned)
Form: AI Scholarship Recommendations	A E V D	A V (own)
Reports: All / Eligible / Approved / Rejected / Pending / Certificate Verification	V E D	—
Report: All Scholarship Details	V E D	V
Page: Scholarship Dashboard.	V	—
Page: ScholarEase Home	V	V
Private fields (Eligibility Status, Review Status, OCR text, AI Recommendation, Document Verification Status)	V E	— (hidden)

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

ScholarEase was created as a new application named **ScholarEase 1** (link name scholarease-1) in the Zoho Creator workspace of the account vishaali.r.2024.cse on 29 October 2026, on the zoho.in data centre. The date and time settings are IST, dd-MMM-yyyy and 24-hour format. The application menu is organised into seven sections: Form 1: Student Details, App Preferences, Scholarship Details, Scholarship Applications, Application Reviews, Scholarship Dashboard. and AI_Scholarship_Recommendations. The default dark navigation theme of Zoho Creator is used, and the ScholarEase Home page provides branding with a title banner and tag line. Figure 5.1 shows the home page with the navigation menu.

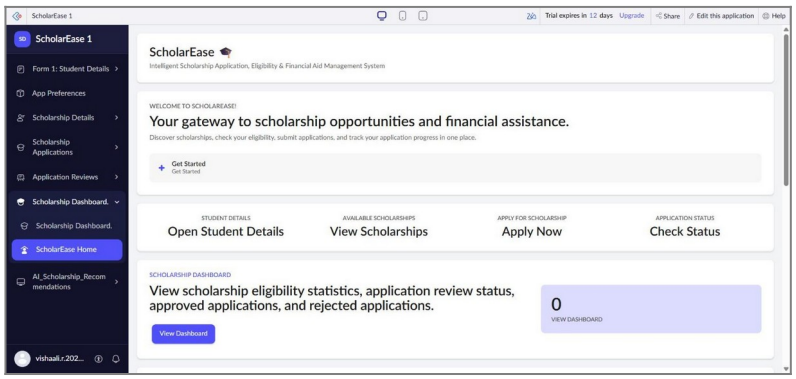


Figure 5.1: ScholarEase Home Page and Navigation Sections

5.2 Forms and Data Capture

Five forms were created as described in Section 4.3. Mandatory fields are enforced by the platform: submitting an empty Student Details form shows an "Invalid entries found" dialog and a message under each mandatory field (Figure 6.1). Each form has a custom success message, for example "Data Added Successfully!" for Student Details, "Scholarship Detail Added Successfully" for Scholarship Details and "Scholarship Application Added Successfully" for applications. Every form has Submit, Update and Cancel actions. Lookups use display formats that show meaningful values: the student lookup shows the first and last name, the scholarship lookup shows the scholarship name and the review lookup shows the Application ID. Status and AI fields are marked private so that only the administrator sees them. No subforms or layout rules are used.

Figure 5.2 shows the forms and their relationships in the Schema Builder, and Figure 5.3 and Figure 5.4 show the main forms in live mode.



Figure 5.2: Forms and Lookup Relationships in the Zoho Creator Schema Builder

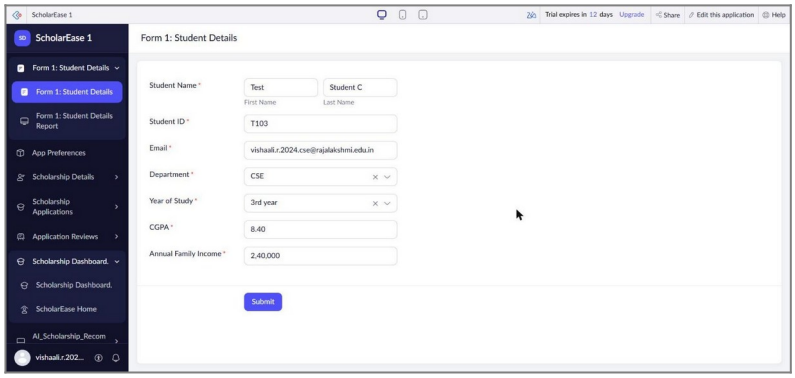


Figure 5.3: Form 1: Student Details Form in Live Mode

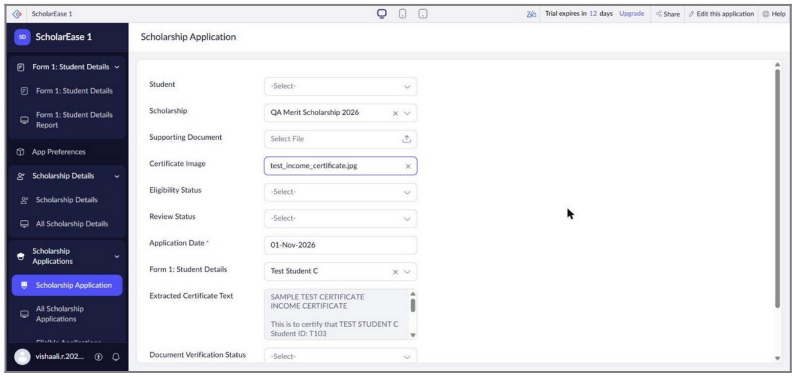


Figure 5.4: Scholarship Application Form in Live Mode with Certificate Image and Extracted Text

5.3 Reports

Ten list reports were created. Filtered reports use the criteria shown in Table 5.1, which are defined in the report definition (for example, show all rows from Scholarship_Application [Review_Status == "Approved"]). All reports provide the standard record menu (Edit, Duplicate, Delete), quick view and detail view layouts, and

column search. Grouping, conditional formatting and custom action buttons are not configured.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Form 1: Student Details Report	List	Form_1_Student_Details	All records	None (standard menu)	Administrator
All Scholarship Details	List	Scholarship_Details	All records	None	Administrator
All Scholarship Applications	List	Scholarship_Application	All records; also embedded in dashboard	None	Administrator
Eligible Applications Report	List	Scholarship_Application	Eligibility_Status == "Eligible"	None	Administrator
Approved Applications Report	List	Scholarship_Application	Review_Status == "Approved"	None	Administrator
Rejected Applications Report	List	Scholarship_Application	Review_Status == "Rejected"	None	Administrator
Pending Reviews Report	List	Scholarship_Application	Review_Status == "Pending"	None	Administrator
Certificate Verification Report	List	Scholarship_Application	All records; shows Certificate Image	None	Administrator
All Application Reviews	List	Application_Review	All records	None	Administrator
AI_Scholarship_Recommendations Report	List	AI_Scholarship_Recommendations	All records	None	Administrator

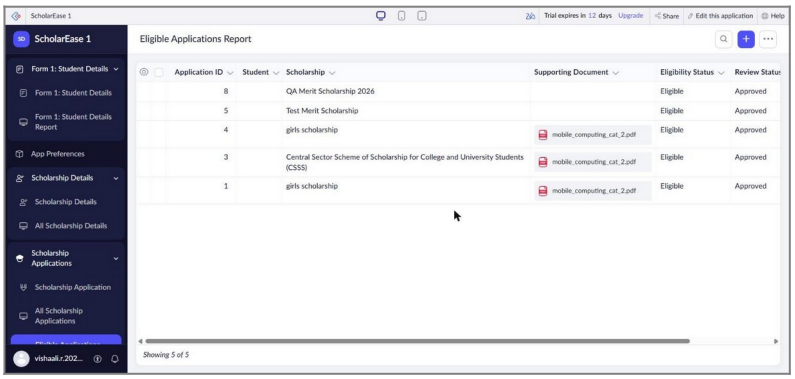


Figure 5.5: Eligible Applications Report (filtered on Eligibility Status)

5.4 Pages and Dashboards

Two pages were built with the Zoho Creator page builder.

ScholarEase Home (target: all users) contains a header panel with the application title and tag line "Intelligent Scholarship Application, Eligibility & Financial Aid Management System", a welcome panel with a Get Started button, four navigation cards

(Student Details, Available Scholarships, Apply for Scholarship, Application Status), a dashboard card with a View Dashboard button, quick links (Explore Scholarships, Submit Application, Track Your Application) and a "ScholarEase AI Assistant" chat panel with a message box, Send and Clear Chat buttons (Figure 5.1).

Scholarship Dashboard. (target: administrator) contains KPI cards for Total Scholarship Application, Eligible Applications, Approved Applications and Rejected Applications; a Scholarship Eligibility Overview pie chart (Eligible / Not Eligible); a Scholarship Review Status Overview pie chart (Approved / Rejected / Additional Documents Required); a second KPI strip; a Monthly Scholarship Application Trends line chart; a Department-wise Applications bar chart; and an embedded All Scholarship Applications report. All values are computed from live records, as verified in test case TC-14. Figure 5.6 shows the KPI cards and status charts.

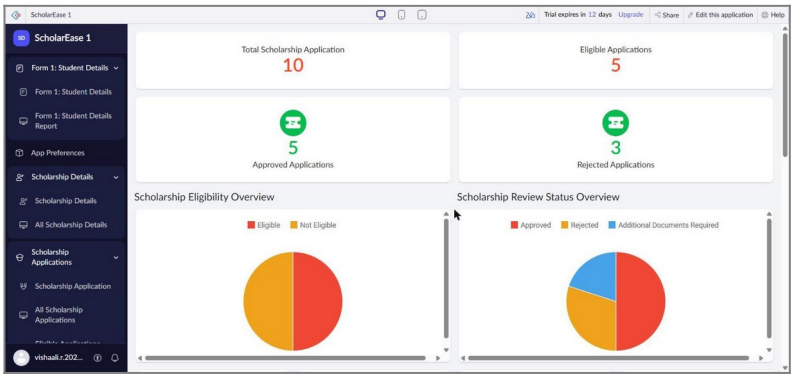


Figure 5.6: Scholarship Dashboard – KPI Cards and Status Charts

5.5 Deluge Scripting

Table 5.2 lists all Deluge scripts in the application. Short, commented snippets are discussed below; the complete code of every script is reproduced in Appendix C.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Scholarship Application → On add → On Validate (Check_Scholarship_Eligibility)	Compute Eligibility Status before the record is saved	fetch by ID, toLong(), count(), null checks, compound if, input field assignment
DS-02	Scholarship Application → On add → On Success (Update_Scholarship_Eligibility)	Re-apply the eligibility rule with a Pending Check default	fetch, nested if-else, variables
DS-03	Scholarship Application → On add → On Success (Application Submitted Email)	Notify administrator mailbox and send confirmation to the student	no-code Send Email task, fetch, sendmail, string concatenation, info

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-04	Application Review → On add → On Success (Update_Application_Review_Status)	Map Decision to Review Status on the linked application	fetch, toString(), trim(), if-else-if, record update
DS-05	Application Review → On add → On Success (Application Approved Email)	E-mail the student when approved	fetch chain, sendmail, zoho.adminuserid
DS-06	Application Review → On add → On Success (Application Rejected Email)	E-mail the student when rejected	fetch chain, sendmail
DS-07	Application Review → On add → On Validate (Additional Documents Email)	E-mail the student when more documents are needed	fetch chain, sendmail, escape sequences
DS-08	AI_Scholarship_Recommendations → On add → On Success (Generate AI Scholarship Recommendation)	Build prompt, call OpenAI, parse and save response	collection fetch, for each, Map, toString(), invokeurl with connection, JSON get(), info, zoho.currentdate
DS-09 to DS-14	Custom functions: Get_Application_Status, Get_Eligibility_Info, Get_scholarship_data, Get_Scholarship_Info, Get_Student_Info, ScholarEase_Chatbot	Return formatted text about applications, eligibility rules, scholarships and the student profile; route chatbot questions	functions with return types and parameters, zoho.loginuserid, for each over forms, string contains(), thisapp.function calls

DS-01 – Eligibility verification. The script runs in the On Validate event, so the status is decided before the record is stored. It fetches the selected student and scholarship records, starts from Not Eligible, and changes the status to Eligible only when all three rules hold.

Code 5.1: Eligibility check (DS-01, Scholarship Application – On Validate)

```
// DS-01 | Form: Scholarship_Application | Event: On add - On Validate
if(input.Form_1_Student_Details != null && input.Scholarship != null &&
input.Application_Date != null)
{
    studentRecord = Form_1_Student_Details[ID ==
input.Form_1_Student_Details.toLong()];
    scholarshipRecord = Scholarship_Details[ID ==
input.Scholarship.toLong()];
    eligibility = "Not Eligible";
    if(studentRecord.count() > 0 && scholarshipRecord.count() > 0)
    {
        // all three conditions must hold
        if(studentRecord.CGPA >= scholarshipRecord.Minimum_CGPA
           && studentRecord.Annual_Family_Income <=
scholarshipRecord.Maximum_Family_Income
           && input.Application_Date <=
scholarshipRecord.Application_Deadline)
        {
            eligibility = "Eligible";
        }
    }
    input.Eligibility_Status = eligibility;
}
```

The full script also checks that every compared value is not null before evaluating the condition (Appendix C.1). DS-02 applies the same rule in the On Success event and uses Pending Check as the starting value when the student, scholarship or date is missing. With the test data of Chapter 6, an application with CGPA 8.40, income ₹2,40,000 and date 01-Nov-2026 against a scheme with minimum CGPA 8.00, maximum income ₹3,00,000 and deadline 30-Nov-2026 was marked Eligible, while a CGPA of 7.10 or a date after the deadline produced Not Eligible.

DS-04 – Review status propagation. When a review is submitted, the decision text is trimmed and mapped to the application's Review Status. Assigning a value to a field of a fetched record updates that record (Appendix C.3).

DS-05 to DS-07 – Decision e-mails. Each decision e-mail workflow follows the chain Application Review → Scholarship Application → Form 1: Student Details to obtain the student's e-mail address and calls sendmail from zoho.adminuserid with a decision-specific subject and message. The approval script is listed in Appendix C.4; the rejection and additional-documents scripts are identical except for the decision condition, the subject ("ScholarEase - Scholarship Application Update" and "ScholarEase - Additional Documents Required") and the message text, and the additional-documents script runs On Validate.

DS-08 – Recommendation processing is described with the integration in Section 5.7. **DS-09 to DS-14** are custom functions in the Default namespace. Get_Student_Info and Get_Application_Status look up the student whose Email equals zoho.loginuserid and return the profile or a list of the student's applications with eligibility and review status; Get_Scholarship_Info and Get_Eligibility_Info list all schemes with their criteria; Get_scholarship_data(student_id) returns the profile of a given student together with the schemes for which the student currently meets all three rules; and ScholarEase_Chatbot(user_message) routes a question to the appropriate function by checking whether it contains words such as "scholarship", "eligibility", "cgpa", "application", "status", "my details" or "my profile".

5.6 Workflows and Automation

All automation is implemented as form workflows. No schedules, approval processes, Blueprints, batch workflows or payment workflows are configured (the corresponding tabs of the Workflow builder are empty). Table 5.3 lists every workflow.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Check_Scholarship_Eligibility	Form workflow	Scholarship Application, record created – On Validate	Student, scholarship and date present	Set Eligibility_Status	DS-01
Update_Scholarship_Eligibility	Form workflow	Scholarship Application, created – On Success	—	Re-evaluate eligibility	DS-02
Application Submitted Email	Form workflow (3 actions)	Scholarship Application, created – On Success	Student linked	Send Email task; log e-mail; sendmail to student	DS-03
Update Student Name Display	Form workflow	Scholarship Application, created – On Success	—	No actions configured	—
Update_Application_Review_Status (1)	Form workflow	Application Review, created – On Success	—	No actions configured	—
Update_Application_Review_Status (2)	Form workflow	Application Review, created – On Success	Application and decision present	Update Review_Status	DS-04
Application Approved Email	Form workflow	Application Review, created – On Success	Decision = Approved	sendmail to student	DS-05
Application Rejected Email	Form workflow	Application Review, created – On Success	Decision = Rejected	sendmail to student	DS-06
Additional Documents Email	Form workflow	Application Review, created – On Validate	Decision = Additional Document Needed	sendmail to student	DS-07
Generate AI Scholarship Recommendation	Form workflow	AI_Scholarship_Recommendations, created – On Success	Student selected	invokeurl to OpenAI; save response	DS-08

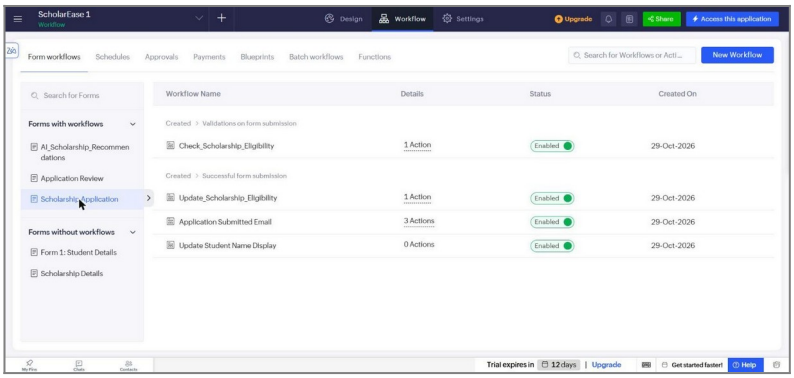


Figure 5.7: Form Workflows Configured on the Scholarship Application Form

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
AI recommendation	OpenAI Responses API – POST https://api.openai.com/v1/responses	invokeurl with connection "openai_connection" (custom connector)	Credentials stored in the Zoho connection; no key in scripts	Sends model gpt-4.1-mini, prompt with student CGPA, income, request and all schemes, max_output_tokens 700; receives JSON output text
Certificate OCR	Zia OCR AI field (Zoho Creator)	AI field with base field Certificate_Image	Platform-internal	Reads the uploaded image and writes visible text to the Extracted Certificate Text field
E-mail	Zoho Creator mail service	sendmail task in Deluge and no-code Send Email task	Platform-internal; from zoho.adminuserid	Submission confirmation and decision notifications
AI agent microservice	ScholarEase AI Recommendation Agent (scholarease_ai_recommendation_agent)	Added under Microservices → AI Models	Platform-internal	Configured in the account; not referenced by the current workflows

Working process of the OpenAI integration (DS-08). When an AI Scholarship Recommendations record is created, the workflow (1) reads the selected student; (2) fetches the student record and takes the CGPA and annual family income; (3) fetches every Scholarship_Details record and concatenates name, type, minimum CGPA, maximum income, amount and deadline into a text block; (4) builds a prompt that instructs the model to recommend suitable scholarships, explain why, mention unmet eligibility conditions, never guarantee approval and use simple language; (5) creates a request map with model, input and max_output_tokens and posts it with invokeurl through the connection; (6) logs the raw response with info; (7) iterates over the output list, picks items of type "message" and concatenates content items of type "output_text"; and (8) writes the text and the current date to the same record, or logs "AI recommendation is empty. Check the API response above." if nothing was returned. The recommendation is advisory: it is stored for the student's information and is not used by the eligibility workflow or the review workflow. The model does not receive student names or e-mail addresses, and no data is used for training by the application.

Code 5.2: OpenAI request and response parsing (DS-08, excerpt)

```
requestBody = Map();
requestBody.put("model","gpt-4.1-mini");
requestBody.put("input",promptText);
requestBody.put("max_output_tokens",700);
apiResponse = invokeurl
[
    url :"https://api.openai.com/v1/responses"
    type :POST
    body:requestBody.toString()
    headers:{"Content-Type":"application/json"}
    connection:"openai_connection"
];
info apiResponse;
for each outputItem in apiResponse.get("output")
{
    if(outputItem.get("type") == "message")
    {
        for each contentItem in outputItem.get("content")
        {
            if(contentItem.get("type") == "output_text")
            { recommendationText = recommendationText +
contentItem.get("text"); }
        }
    }
}
```

During testing the connection was in the Connected state (Figure 5.8), but the API returned an insufficient_quota error because the linked OpenAI account had no remaining credits; the effect of this on the results is reported in Section 6.2 and Section 6.3. No Postman test was performed; the request and response were inspected through the Deluge info output instead.

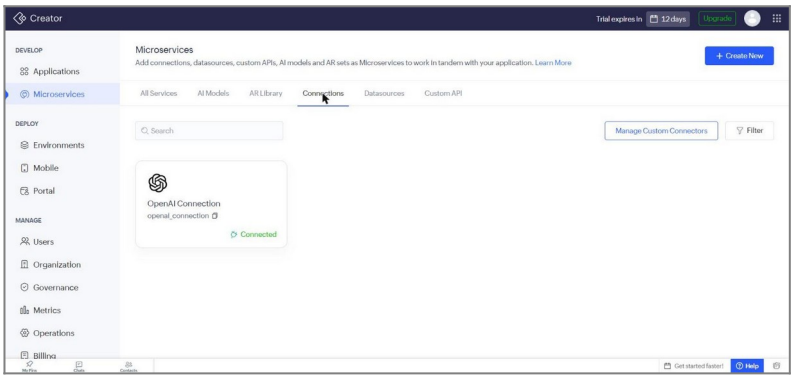


Figure 5.8: OpenAI Connection (openai_connection) in Microservices → Connections

5.8 User Management and Role-Based Access

The application currently has a single user, the application owner, who acts as the administrator and has full access to every form, report, page and setting. The User Permissions page contains only the two default permission profiles provided by Zoho Creator – Read (read permission for all components) and Write (write permission for all components) – and no additional users, custom roles, record-level criteria or customer

portal users have been configured. Field-level protection is provided by marking Eligibility Status, Review Status, Extracted Certificate Text, AI Recommendation and Document Verification Status as private fields, and Student Name and Email are flagged as personal data. The custom functions Get_Student_Info and Get_Application_Status already filter records by zoho.loginuserid, which prepares the application for future student accounts with record-level access as planned in Table 4.6.

5.9 AI and Analytics Features

OCR (Zia AI field). The field Extracted Certificate Text (link name OCR) is a Zia AI field of type OCR whose base field is Certificate_Image. When an image is uploaded in the Scholarship Application form, the platform processes it and displays the extracted text in the form before the record is submitted. In test case TC-09 a certificate image containing the lines "SAMPLE TEST CERTIFICATE", "INCOME CERTIFICATE", "This is to certify that TEST STUDENT C" and "Student ID: T103" was uploaded and the same lines appeared in the field (Figure 5.4). OCR is used only to help the administrator read certificate details; it does not establish whether a certificate is genuine, and the Document Verification Status field is set by the administrator.

Large language model recommendation. The OpenAI integration described in Section 5.7 is the generative-AI feature of the application. Its output is advisory and is never used for approval. **Analytics.** Analytics are provided by the charts and KPI cards on the Scholarship Dashboard page (Section 5.4); Zoho Analytics and pivot reports are not used. **AI-generated content.** The OCR text and the recommendation text are produced by AI services at run time. All Deluge scripts were reviewed and verified through the test cases in Chapter 6.

5.10 Pro-Code Extension

The application does not use Zoho Catalyst, JavaScript widgets or the Creator REST API; its extension beyond the visual builders is limited to Deluge custom functions (DS-09 to DS-14) and the invokeurl call to the OpenAI REST API, which was sufficient for the rule-based scholarship process. A Catalyst function for batch recommendations, a JavaScript widget for an interactive chat panel, or the Creator REST API for ERP integration would be the natural pro-code extensions.

5.11 Mobile and Portal Access

No customer portal is configured. The application's forms are available in the Zoho Creator mobile layout without extra development, as shown in the mobile preview of the Student Details form in Figure 5.9. No mobile-specific layout or offline settings were configured.

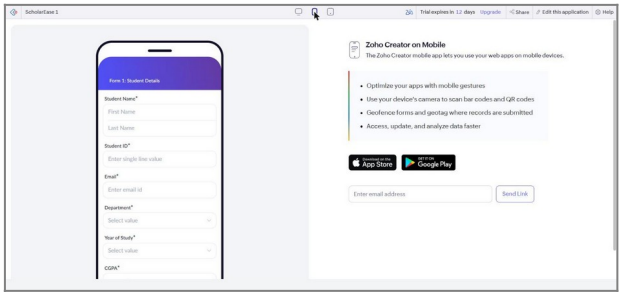


Figure 5.9: ScholarEase in the Zoho Creator Mobile Preview

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out on the live application in the development environment on 01 November 2026 using the administrator account. Dedicated test records were created so that existing data was not modified: two students (Test Student C – T103, CSE, 3rd year, CGPA 8.40, income ₹2,40,000; Test Student D – T104, ECE, 2nd year, CGPA 7.10, income ₹2,00,000), two scholarships (QA Merit Scholarship 2026 – Merit-Based, minimum CGPA 8.00, maximum income ₹3,00,000, amount ₹15,000, deadline 30-Nov-2026; QA Expired Scholarship – Need-Based, minimum CGPA 7.00, maximum income ₹5,00,000, amount ₹5,000, deadline 31-Oct-2026) and a test certificate image. The test students use the developer's own college e-mail address.

- **Unit testing of forms:** each form was submitted with valid data and with missing mandatory fields.
- **Unit testing of scripts:** the eligibility rule was tested with an eligible case, an ineligible CGPA case and a deadline case; each review decision was tested once.
- **Integration testing:** the OCR field, the OpenAI connection and the e-mail workflows were exercised end to end.
- **System testing:** report filters and dashboard KPIs were compared with the underlying records before and after the test data was added.
- **Role-based access testing** was not performed because only the administrator account exists (Section 5.8).

6.2 Test Cases

Table 6.1 lists the sixteen test cases executed. Application IDs 8, 9 and 10 were created during testing.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a student with all mandatory fields	Test Student C, T103, e-mail, CSE, 3rd year, 8.40, 2,40,000	Record created; success message	"Data Added Successfully!" shown; record listed in report	Pass
TC-02	FR-01	Submit Student Details with mandatory fields empty	Click Submit on empty form	Validation error; record not saved	"Invalid entries found" dialog and field messages shown	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-03	FR-02	Create scholarships	QA Merit Scholarship 2026; QA Expired Scholarship	Records created	"Scholarship Detail Added Successfully" shown; both listed	Pass
TC-04	FR-03	Submit an application with certificate image	Test Student C + QA Merit, 01-Nov-2026, image	Record created with Application ID	"Scholarship Application Added Successfully"; Application ID 8	Pass
TC-05	FR-04	Eligibility – eligible student	App 8: 8.40 ≥ 8.00; 2,40,000 ≤ 3,00,000; 01-Nov ≤ 30-Nov	Eligible	Eligible	Pass
TC-06	FR-04	Eligibility – CGPA below minimum	App 9: Test Student D (7.10) + QA Merit	Not Eligible	Not Eligible	Pass
TC-07	FR-04	Deadline validation	App 10: Test Student C + QA Expired (deadline 31-Oct-2026), date 01-Nov-2026	Not Eligible	Not Eligible	Pass
TC-08	FR-06	OpenAI recommendation	Test Student C, standard request text	AI Recommendation and date filled	Record saved; AI Recommendation empty; log shows insufficient_quota error from OpenAI	Fail
TC-09	FR-05	OCR certificate text extraction	Upload test income certificate image	Visible text in Extracted Certificate Text	All certificate lines extracted correctly	Pass
TC-10	FR-07	Application approval	Review of App 8 – Approved	Review Status = Approved	Approved	Pass
TC-11	FR-07	Application rejection	Review of App 9 – Rejected	Review Status = Rejected	Rejected	Pass
TC-12	FR-07	Additional document request	Review of App 10 – Additional Document Needed	Review Status = Additional Documents Required	Additional Documents Required	Pass
TC-13	FR-08	Application submission e-mail	Submission of App 8	Confirmation e-mail with Application ID	E-mail "ScholarEase - Application Submitted Successfully" received in the student mailbox	Pass
TC-14	FR-10	Dashboard KPI calculation	Compare KPIs before and after tests	KPIs equal record counts	Before 7/4/4/2, after 10/5/5/3 (total/eligible/approved/rejected), matching reports	Pass
TC-15	FR-11	Report filtering and display	Open Eligible, Approved, Rejected and Pending reports	Only matching records shown	Eligible 5 (IDs 8,5,4,3,1); Approved 5; Rejected 3 (9,6,2); Pending 0	Pass
TC-16	FR-09	Decision e-mails	Reviews of Apps 8, 9 and 10	Approval, rejection and additional-documents e-mails	Approved, Application Update (rejection) and Additional Documents Required e-mails received	Pass

Observations recorded during testing. (1) Newly submitted applications have no initial Review Status, so they do not appear in the Pending Reviews Report, whose filter is Review_Status == "Pending", until a review is recorded (Figure 6.2 shows Applications 8–10 before review). (2) The Additional Documents Email workflow runs in the On Validate event of the review form, whereas the other decision e-mails run On Success. (3) The confirmation e-mail inserts input.ID, the internal record ID, rather than

the auto-numbered Application ID. (4) Two workflows have no actions. (5) The ScholarEase AI Assistant panel on the home page did not display a reply when a question was sent. These observations are recorded for future improvement and did not affect the other test results.

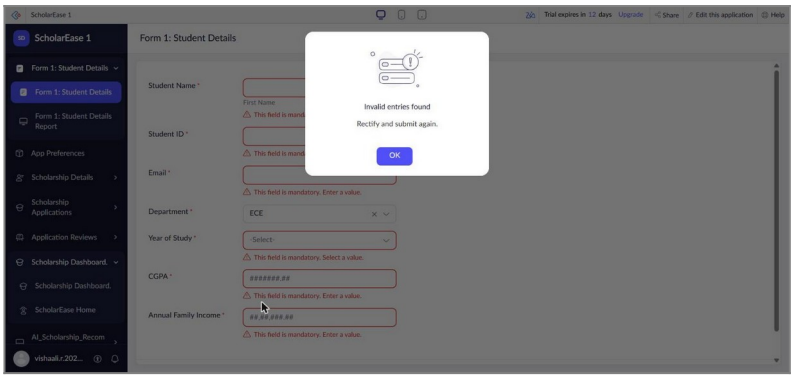


Figure 6.1: TC-02 – Mandatory Field Validation on Form 1: Student Details

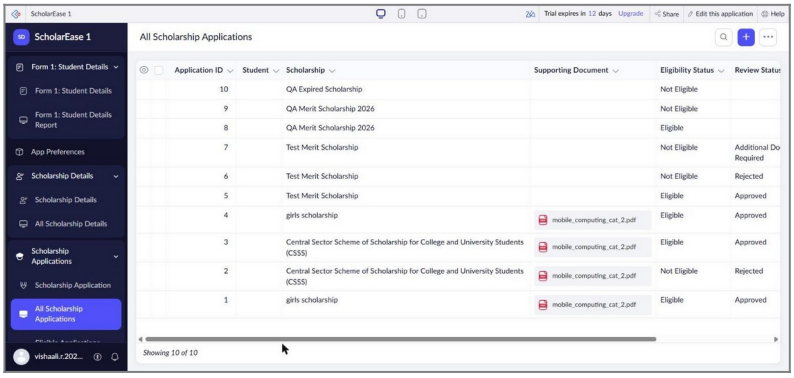


Figure 6.2: TC-04 to TC-07 – Eligibility Status of Applications 8, 9 and 10 before Review

6.3 Debugging Deluge Scripts

Scripts were debugged with Deluge info statements, whose output appears in the "View Log Details" link that Zoho Creator shows after a form is submitted, and by reading scripts in the Application IDE. The recommendation workflow logs the raw API response with info apiResponse and logs an explicit message when no text could be extracted. During TC-08 this log showed the response {"error":{"message":"You have no credits remaining. Add credits to continue using the API ...","type":"insufficient_quota","param":null,"code":"credit_balance_exhausted"} } followed by "AI recommendation is empty. Check the API response above." (Figure 6.3). The log confirms that the script, the connection and the request reached the OpenAI service successfully and that the failure is caused by the account's credit balance, not by the Deluge code. The fix is to add credits to the OpenAI account or to link the connection to an account with available quota; no code change is required.

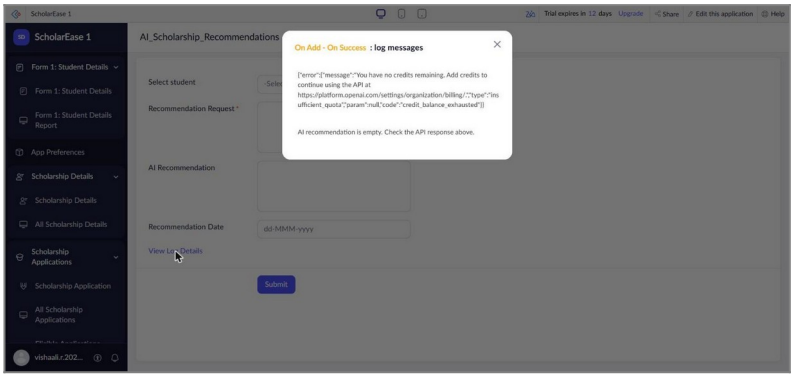


Figure 6.3: Deluge Log Output of the AI Recommendation Workflow (TC-08)

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments for multi-stage deployment. ScholarEase is currently maintained only in the Development environment: the Environments page shows that no application has been added to Stage or Production (Figure 7.1), and no version has been published. The application is used directly through its development URL.

The planned strategy is to add the application to Environments, publish to Stage, repeat the Chapter 6 tests, authorise a funded OpenAI connection and student users, publish to Production, and keep the previous version in Version History for rollback.

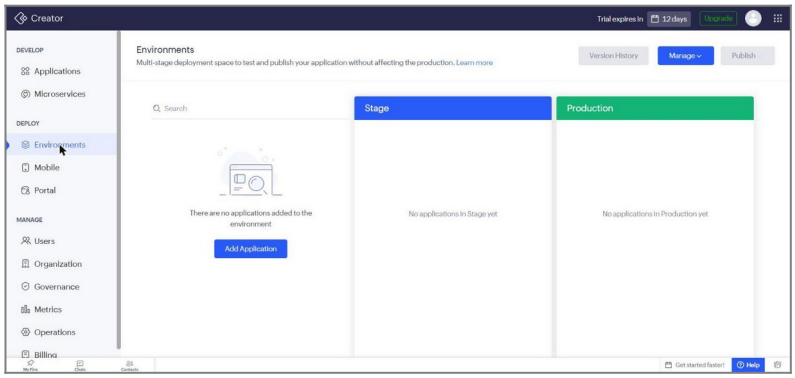


Figure 7.1: Environments Page – No Applications in Stage or Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (√)
1	All test cases executed in Development / Stage	√ (Development)
2	Sample/test data removed or isolated	✗ (test records retained for demonstration)
3	Roles, permissions and users configured in Production	✗
4	Connections authorised in Production	✗
5	Schedules and workflows enabled	√ (all form workflows enabled; no schedules)
6	Application published and URL verified	✗
7	Version / release notes recorded	✗

7.2 Performance Optimisation

The scripts use criteria-based fetch statements by record ID (for example Scholarship_Details[ID == input.Scholarship]) so that only the required records are read, and filtered reports restrict the records loaded for each status view.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account authentication	Access to the application and builder	Unauthorised access
Connection-based credentials	OpenAI call (connection openai_connection)	Leakage of the API key in scripts
Private fields	Eligibility Status, Review Status, Extracted Certificate Text, AI Recommendation, Document Verification Status	Students altering or viewing decision fields
Personal data flags	Student Name and Email	Uncontrolled handling of personal data
Mandatory field validation	All forms	Incomplete or invalid records
Data minimisation in AI prompt	Recommendation workflow sends only CGPA, income, request text and scheme data	Disclosure of student identity to an external service
Human approval	Application Review form	Automated approval based on AI output

Two-factor authentication, IP restrictions, encrypted fields and custom security policies (Governance → Security Policies) are not configured.

7.4 Governance and Administration

Workflows and functions use readable link names (for example Check_Scholarship_Eligibility) and can be reviewed and exported from the Application IDE, and the Schema Builder documents the form relationships. Backup, Audit Trail and Logs administration under Operations have not been configured;

because the Logs view returned no entries during testing, e-mail delivery was confirmed in the recipient mailbox. Administration is the responsibility of the application owner.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is demonstrated below by following one test scenario through each step: a student is registered, an application is submitted with a certificate image, the eligibility result is produced, an administrator records a decision, and the dashboard is updated.

Step 1 – Student registration. Test Student C was registered and the success message "Data Added Successfully!" was displayed (Figure 5.3).

Step 2 – Application with certificate image. The student applied for QA Merit Scholarship 2026 and uploaded a certificate image; the OCR field displayed the extracted text (Figure 5.4) and the application was saved as Application ID 8 with Eligibility Status Eligible.

Step 3 – Review decision. The administrator recorded "Approved" for Application 8 in the Application Review form, and the Review Status of the application changed automatically (Figure 8.1).

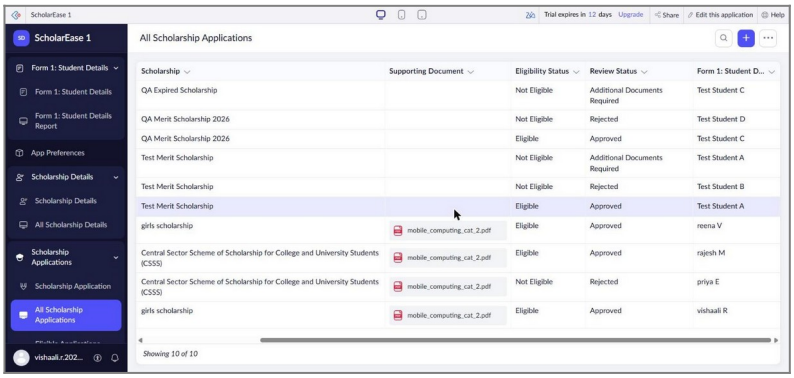


Figure 8.1: Scenario Step 3 – Review Status Updated on Applications 8, 9 and 10

Step 4 – Dashboard update. The dashboard KPI cards changed from 7 total / 4 eligible / 4 approved / 2 rejected before the scenario to 10 / 5 / 5 / 3 afterwards (Figure 5.6).

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Data model and forms	Sections 4.2, 4.3, 5.2	Five linked forms in use; TC-01 to TC-04 passed
Automatic eligibility verification	Sections 5.5, 5.6 (DS-01, DS-02)	TC-05 to TC-07 passed
Review propagation and notifications	Sections 5.5, 5.6 (DS-03 to DS-07)	TC-10 to TC-13 and TC-16 passed
OpenAI and OCR integration	Sections 5.7, 5.9 (DS-08, OCR field)	OCR achieved (TC-09); OpenAI integration implemented but blocked by API credit balance (TC-08)
Reports and dashboard	Sections 5.3, 5.4	TC-14 and TC-15 passed
Secure production deployment	Chapter 7	Partially achieved: security controls in place; Production deployment pending

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Eligibility check	Manual comparison of CGPA, income and date for each application	Computed automatically on submission	No manual calculation; consistent results
Manual steps	Form filling, register entry, manual checks, retyping certificate data, manual notification	Single form submission; OCR text; automatic status and e-mail	Several manual steps removed
Report generation	Manual counting at the end of the cycle	Live filtered reports and dashboard KPIs	Real-time statistics
Data accessibility	Paper files in the office	Central online records accessible from browser and mobile layout	Anytime access for authorised users
Status tracking	Enquiry at the office	Eligibility and review status on every record	Transparent status

8.4 Limitations of the Low-Code Approach

Several constraints of the platform and of the current build were observed. **External AI dependence:** the recommendation feature depends entirely on the availability and quota of the OpenAI account; when the credit balance was exhausted the feature stopped working although the low-code logic was correct. **AI and OCR quality:** recommendation quality depends on the completeness of the student and scholarship data supplied in the prompt, and OCR accuracy depends on image quality and clarity;

OCR does not verify authenticity. **User-interface limits:** page layouts are restricted to builder components, and the interactive assistant panel on the home page requires additional scripting to become functional. **Trial and licensing:** the application runs on a time-limited trial, and user licences and AI usage are subject to plan limits. **Access configuration:** meaningful separation between students and administrators requires additional users, permission profiles and record-level criteria. **Vendor lock-in:** forms, workflows and Deluge scripts are specific to Zoho Creator; the Application IDE export reduces but does not remove this dependence. Final scholarship approval always requires administrative review.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

No-code: the five forms, field types and validations, lookup relationships, the ten list reports with filter criteria, the two pages with KPI cards and charts, the Zia OCR AI field and the no-code Send Email task were configured entirely through visual builders. **Low-code:** eligibility verification, review-status propagation, decision e-mails, the OpenAI request and response handling and the six custom functions required Deluge scripts. These scripts are short and declarative, but they still require an understanding of records, conditions and JSON. **Pro-code:** a fully interactive chat assistant, bulk recommendation processing with caching, ERP integration and advanced document authenticity checks would require JavaScript widgets, Zoho Catalyst functions or external services. The project showed that low-code is the right level for rule-based administrative workflows, and that a move to pro-code is justified only when user-interface interactivity, heavy processing or deep integration are required.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

ScholarEase is an intelligent scholarship management system built on Zoho Creator. Using lookup-based forms, Deluge workflows, a Zia OCR AI field and an OpenAI integration, it centralises student and scholarship records and automates eligibility verification, certificate text extraction, review-status updates, e-mail notifications, reports and a live dashboard.

Live testing confirmed that all modules work as designed except the OpenAI recommendation, which was blocked by the API account's credit balance. The application currently runs in the development environment and keeps final decisions with human reviewers.

9.2 Future Enhancements

The following enhancements are proposed; none is implemented in the current version.

1. Integration with the college ERP and other institutional systems to import student records automatically.
2. Advanced document authenticity verification and automated document completeness checking.
3. A student self-service portal with separate student accounts, permission profiles and record-level access.
4. Advanced scholarship matching, scheduled deadline reminders and additional channels such as SMS.
5. Historical analytics and a multilingual chatbot built on ScholarEase_Chatbot.
6. Publishing to Stage and Production with a funded OpenAI connection.

9.3 Learning Outcomes and Career Relevance

The project built practical skills in modelling processes as related forms, writing Deluge workflows, integrating REST APIs through connections, using AI fields and live testing – skills that map directly to low-code developer and automation engineer roles.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	ScholarEase 1 – seven menu sections	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	12 field types across five forms	5.2	Tables 4.1–4.5
II	Lookup field (relationship between forms)	Y	Scholarship Application, Application Review, AI Recommendations	4.2, 5.2	Fig. 4.2, 5.2
II	Subform (parent–child data)	N	—	4.2, 5.2	—
II	Field validations / layout rules	Partial	Mandatory and format validation; no layout rules	5.2	Fig. 6.1
II	List report with filter, grouping, custom action	Partial	Filtered list reports; no grouping or custom actions	5.3	Table 5.1, Fig. 5.5
II	Kanban / Calendar / Timeline / Map report	N	—	5.3	—
II	Pivot table / chart report	N	Charts are on the dashboard page instead	5.3	—
II	Page / dashboard built with UI Builder	Y	ScholarEase Home, Scholarship Dashboard.	5.4	Fig. 5.6
II	Deluge – form events (validate / success / field actions)	Y	On Validate and On Success scripts	5.5	Code 5.1
II	Deluge – functions, fetch, loops, collections	Y	Six custom functions; for each; Map	5.5	Appendix C
II	Email / in-app notifications	Y	Submission and decision e-mails	5.5, 5.6	Appendix C.2, C.4
II	Form workflow	Y	10 form workflows	5.6	Fig. 5.7
II	Scheduled workflow	N	—	5.6	—
II	Approval process	N	Review handled through the Application Review form	5.6	—
II	Blueprint	N	—	5.6	—
II	Batch / report workflow	N	—	5.6	—
II	Testing & debugging evidence	Y	16 test cases; Deluge log	6	Table 6.1, Fig. 6.3
III	Integration with a Zoho service	Y	Zia OCR AI field; Zoho Creator mail	5.7	Fig. 5.4
III	Connection / REST API call (invokeurl)	Y	openai_connection – OpenAI Responses API	5.7	Code 5.2, Fig. 5.8
III	Webhook or Creator REST API	N	—	5.7	—
III	Users, roles and permission profiles	N	Default profiles only	5.8	Section 5.8
III	Record-level access criteria / customer portal	N	—	5.8	—
III	Zia / AI feature	Y	OCR AI field; OpenAI recommendation	5.9	Fig. 5.4

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts and KPI cards	5.9	Fig. 5.6
IV	Performance optimisation applied & measured	N	Not measured	7.2	—
IV	Deployment across environments	N	Development only	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Connection credentials, private fields, personal-data flags	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Not configured	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	Partial	Deluge custom functions only	5.10	Appendix C
V	Low-code limitations & spectrum analysis	Y	—	8.4, 8.5	—
—	Mobile app / portal access	Partial	Mobile layout preview; no portal	5.11	Fig. 5.9

Total features implemented: 14 / 32 (fully implemented; a further 5 are partially implemented)

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C – DELUGE SOURCE CODE OF WORKFLOWS

The scripts below are transcribed from the Application IDE of ScholarEase 1. The two form workflows without actions (Update Student Name Display and the first Update_Application_Review_Status) contain no script. Update_Scholarship_Eligibility (DS-02) repeats the rule of C.1 with a Pending Check default, the rejection and additional-documents e-mail scripts (DS-06, DS-07) follow C.4, and the six custom functions (DS-09 to DS-14) are short read-only text builders described in Section 5.5; these are not reproduced here.

C.1 Scholarship Application – Check_Scholarship_Eligibility

```
// DS-01 | Scholarship_Application → On add → On Validate | Compute Eligibility Status before
save
if(input.Form_1_Student_Details != null && input.Scholarship != null &&
input.Application_Date != null)
{
    studentRecord = Form_1_Student_Details[ID == input.Form_1_Student_Details.toLong()];
    scholarshipRecord = Scholarship_Details[ID == input.Scholarship.toLong()];
    eligibility = "Not Eligible";
    if(studentRecord.count() > 0 && scholarshipRecord.count() > 0)
    {
        if(studentRecord.CGPA != null && studentRecord.Annual_Family_Income != null &&
scholarshipRecord.Minimum_CGPA != null && scholarshipRecord.Maximum_Family_Income != null &&
scholarshipRecord.Application_Deadline != null)
        {
            if(studentRecord.CGPA >= scholarshipRecord.Minimum_CGPA &&
studentRecord.Annual_Family_Income <= scholarshipRecord.Maximum_Family_Income &&
input.Application_Date <= scholarshipRecord.Application_Deadline)
            {
                eligibility = "Eligible";
            }
        }
    }
    input.Eligibility_Status = eligibility;
}
```

C.2 Scholarship Application – Application Submitted Email

```
// DS-03 | Scholarship_Application → On add → On Success | Administrator notification and
student confirmation e-mail
// Action 1 (no-code "Send Email" task configured in the workflow builder)
//   From    : vishaali.r.2024.cse@rajalakshmi.edu.in
//   To      : vishaalimoni05@gmail.com (administrator mailbox)
//   Subject : "Record ScholarEase - New Scholarship Application ReceivedAdded"
//   Message : "Dear Student, Your scholarship application has been submitted successfully
through
//             ScholarEase. Your application will be reviewed by the scholarship
administration team.
//              Please keep your application details for future reference. Thank you for using
//             ScholarEase! Regards," (rich-text body, content type Plain Text, default
template inline)
// Action 2 (custom Deluge script) - retrieves and logs the student's e-mail address
if(input.Form_1_Student_Details != null)
{
    studentRecord = Form_1_Student_Details[ID == input.Form_1_Student_Details];
    if(studentRecord.ID != null && studentRecord.Email != null)
    {
        studentEmail = studentRecord.Email;
        info studentEmail;
    }
}
// Action 3 (custom Deluge script) - confirmation e-mail to the student
if(input.Form_1_Student_Details != null)
{
    studentRecord = Form_1_Student_Details[ID == input.Form_1_Student_Details];
    if(studentRecord.ID != null && studentRecord.Email != null)
    {
        sendmail
        [
            from :zoho.adminuserid
            to :studentRecord.Email
            subject :"ScholarEase - Application Submitted Successfully"
            message :"Dear Student, your scholarship application has been submitted
successfully. Your Application ID is " + input.ID + ". Please log in to ScholarEase to check
your application status."
        ]
    }
}
```

C.3 Application Review – Update_Application_Review_Status

```
// DS-04 | Application_Review → On add → On Success | Propagate decision to Review Status
if(input.Application != null && input.Decision != null)
{
    applicationID = input.Application;
    decisionText = input.Decision.toString().trim();
    if(decisionText == "Approved")
    {
        applicationRecord = Scholarship_Application[ID == applicationID];
        applicationRecord.Review_Status="Approved";
    }
    else if(decisionText == "Rejected")
    {
        applicationRecord = Scholarship_Application[ID == applicationID];
        applicationRecord.Review_Status="Rejected";
    }
    else if(decisionText == "Additional Document Needed")
    {
        applicationRecord = Scholarship_Application[ID == applicationID];
        applicationRecord.Review_Status="Additional Documents Required";
    }
}
```

C.4 Application Review – Application Approved Email

```
// DS-05 | Application_Review → On add → On Success | Approval e-mail
if(input.Application != null && input.Decision == "Approved")
{
    applicationRecord = Scholarship_Application[ID == input.Application];
    if(applicationRecord.ID != null && applicationRecord.Form_1_Student_Details != null)
    {
        studentRecord = Form_1_Student_Details[ID == applicationRecord.Form_1_Student_Details];
        if(studentRecord.ID != null && studentRecord.Email != null)
        {
            sendmail
            [
                from :zoho.adminuserid
                to :studentRecord.Email
                subject :"ScholarEase - Scholarship Application Approved"
                message :"Congratulations! Your scholarship application has been approved.
Please log in to ScholarEase to view your application details."
            ]
        }
    }
}
```

C.5 AI_Scholarship_Recommendations – Generate AI Scholarship Recommendation

```
// DS-08 | AI_Scholarship_Recommendations → On add → On Success | OpenAI recommendation
studentID = input.AI_Scholarship_Recommendations;
if(studentID != null)
{
    studentRecord = Form_1_Student_Details[ID == studentID];
    if(studentRecord.ID != null)
    {
        // Fetch student information
        studentCGPA = studentRecord.CGPA;
        studentIncome = studentRecord.Annual_Family_Income;
        // Fetch available scholarships
        scholarshipText = "";
        scholarshipList = Scholarship_Details[ID != 0];
        for each  scholarship in scholarshipList
        {
            scholarshipText = scholarshipText + "\nScholarship: " +
scholarship.Scholarship_Name + "\nType: " + scholarship.Scholarship_Type + "\nMinimum CGPA: " +
scholarship.Minimum_CGPA + "\nMaximum Family Income: " + scholarship.Maximum_Family_Income + "\
nAmount: " + scholarship.Scholarship_Amount + "\nDeadline: " + scholarship.Application_Deadline
+ "\n";
        }
        // Build AI prompt
        promptText = "You are a scholarship recommendation assistant." + "\nStudent CGPA: " +
studentCGPA + "\nAnnual Family Income: " + studentIncome + "\nStudent Request: " +
input.Recommendation_Request + "\nAvailable Scholarships: " + scholarshipText + "\nRecommend
suitable scholarships based on the student's profile." + "\nExplain why each scholarship is
suitable." + "\nMention unmet eligibility conditions." + "\nNever guarantee scholarship
approval." + "\nUse simple and understandable language.";
        // Prepare OpenAI request
        requestBody = Map();
        requestBody.put("model","gpt-4.1-mini");
        requestBody.put("input",promptText);
        requestBody.put("max_output_tokens",700);
        // Call OpenAI API
        apiResponse = invokeurl
        [
            url :"https://api.openai.com/v1/responses"
            type :POST
            body:requestBody.toString()
            headers:{"Content-Type":"application/json"}
            connection:"openai_connection"
        ];
        info apiResponse;
        recommendationText = "";
        // Extract AI response
        if(apiResponse != null)
        {
            outputList = apiResponse.get("output");
            if(outputList != null)
            {
```

```
                for each  outputItem in outputList
                {
                    if(outputItem.get("type") == "message")
                    {
                        contentList = outputItem.get("content");
                        if(contentList != null)
                        {
                            for each  contentItem in contentList
                            {
                                if(contentItem.get("type") == "output_text")
                                {
                                    recommendationText = recommendationText +
contentItem.get("text");
                                }
                            }
                        }
                    }
                }
            }
        }
        // Save AI recommendation
        if(recommendationText != "")
        {
            recommendationRecord = AI_Scholarship_Recommendations[ID == input.ID];
            recommendationRecord.AI_Recommendation=recommendationText;
            recommendationRecord.Recommendation_Date=zoho.currentdate;
            info "AI recommendation saved successfully.";
        }
        else
        {
            info "AI recommendation is empty. Check the API response above.";
        }
    }
}
```

APPENDIX D – USER MANUAL

Access. Open https://creatorapp.zoho.in/vishaali.r.2024.cse_rajal188/scholarease-1 in a web browser (or the Zoho Creator mobile app) and sign in with an authorised Zoho account. ScholarEase Home is available under the Scholarship Dashboard. section.

Table D.1: User Manual – Common Tasks

User	Task	Menu Path and Steps
Administrator	Add a scholarship	Scholarship Details → Scholarship Details; enter name, type, minimum CGPA, maximum income, amount and deadline; Submit.
Administrator / Student	Register a student	Form 1: Student Details → Form 1: Student Details; fill all mandatory fields; Submit.
Student	Apply for a scholarship	Scholarship Applications → Scholarship Application; select scholarship and student, upload certificate image and document, check date; Submit. Eligibility is computed automatically.
Student	Request a recommendation	AI_Scholarship_Recommendations → AI_Scholarship_Recommendations; select student, type the question; Submit.
Administrator	Review an application	Application Reviews → Application Review; select Application ID, enter reviewer and date, choose the decision; Submit. Status and e-mail update automatically.
Administrator	Verify certificates	Scholarship Applications → Certificate Verification Report; compare image with extracted text; set Document Verification Status.
Administrator	Monitor statistics	Scholarship Dashboard. → Scholarship Dashboard.; filtered reports under Scholarship Applications.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	VISHAALI R	Register No.	2116240701595
Project Title	ScholarEase – Intelligent Scholarship Application, Eligibility and Financial Aid Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner **Signature of External Examiner**

Name: _____ Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	VISHAALI R	Register No.	2116240701595
Department / Section	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

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_____

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____